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Late Cretaceous paleomagnetic data from the Median Range of Kamchatka, Russia: tectonic implications

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Abstract

A Late Cretaceous island arc complex was sampled for paleomagnetic study in the northern part of the Median Range of Kamchatka. In most samples, thermal demagnetization in 12–18 steps revealed either a reversed characteristic remanent magnetization (ChRM) or a remagnetization circle. Combining these yielded a positive fold test ($K_s/K_g = 5.8$) and we interpreted the ChRM to be primary. The tilt corrected overall mean direction of dec. = 105.5°, inc. = −67.0°, $\alpha_{95} = 4.1°$ for these rocks corresponds to a paleolatitude of 49.7° ± 5.6°N. This value is 20° to 22° lower than the Late Cretaceous North American reference values, and northward transport of the subject tectonostratigraphic terrane and island arc to which it had originally belonged, is strongly indicated. The observed paleolatitudes are approximately similar for all Late Cretaceous island arc complexes in Kamchatka and the southern part of the Koryak Highlands. We conclude that these complexes were originally part of the same island arc which originated 20° to 25° south of the continental margin in the Campanian, and then was transported northward until its emplacement in the Paleocene. Northward drift of the island arc coincided with its activity, and the related subduction zone also had to drift with a rapidly moving oceanic plate, most probably the Kula plate. This relationship implies that the subduction zone dipped southward and was consuming oceanic lithosphere, presumably the oceanic periphery of the North American plate. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Kamchatka Peninsula; Cretaceous; paleomagnetism; kinematics

1. Introduction

It is now widely recognized that broad region associated with continental masses surrounding the North Pacific are an agglomerate of terranes. Many of these terranes are thought to be remnants of intraoceanic island arcs transported from far away and subsequently accreted to the adjacent continents. Such terranes comprise most of the Kamchatka Peninsula and the Olutor zone of the Koryak Highlands [1–3].

Late Cretaceous volcanoclastic complexes are known in Kamchatka mainly in the Median Range and the East Ranges (Fig. 1). The structures of the East Ranges can be traced to the north from Kamchatka to Karaginsky Island. Further to the north, Late Cretaceous arc-related complexes are widespread in the Olutor zone, the southern part of the Koryak Highlands (Fig. 1). In the Median Range, a Late Cretaceous island-arc complex is thrust over

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Campanian–Maastrichtian terrigenous formations of the West zone of Kamchatka along the Lesnov thrust. The Median Range and East Ranges tectonic units are separated by the sediment-filled Late Cenozoic Central Kamchatka basin. Geochemically, the Late Cretaceous volcanoclastic complexes from both the Median Range and the East Ranges are similar to volcanic rocks of intraoceanic island arcs such as the Tonga arc [4,5].

Numerous attempts to reconstruct the tectonic evolution of the convergent boundary of the northwest Pacific have been undertaken [2,3,5]. To substantiate such reconstructions, one should know the duration of volcanic activity of an island arc, its original position, and the age of deformation during its collision with the continent. Paleomagnetic data, combined with a plate tectonic model, can provide quantitative estimations of original positions and subsequent horizontal motions of the island arcs. Paleomagnetic data on the Late Cretaceous island-arc complexes have already been obtained from the northern part of the Eastern Ranges [6], Karaginsky Island [7] and Ilpinsky Peninsula [8]. Also, the data on similar complexes from the Olutor zone [9] are available. These results are based on fully demagnetized collections and confirmed by the field tests. According to these paleomagnetic data, the studied Late Cretaceous island-arc complexes were formed approximately 20–25° south of the continental margin.

The Median Range and East Ranges complexes are sometimes regarded as the remnants of one island arc [2]. The available geological and paleomagnetic data, however, are not sufficient to exclude the possibility that two co-existing in the Campanian–Maastrichtian active island arcs were later tectonically juxtaposed. If the Median Range and East Ranges island-arc complexes had originally belonged to the same island arc, their paleolatitudes should agree; otherwise, they should differ. To test

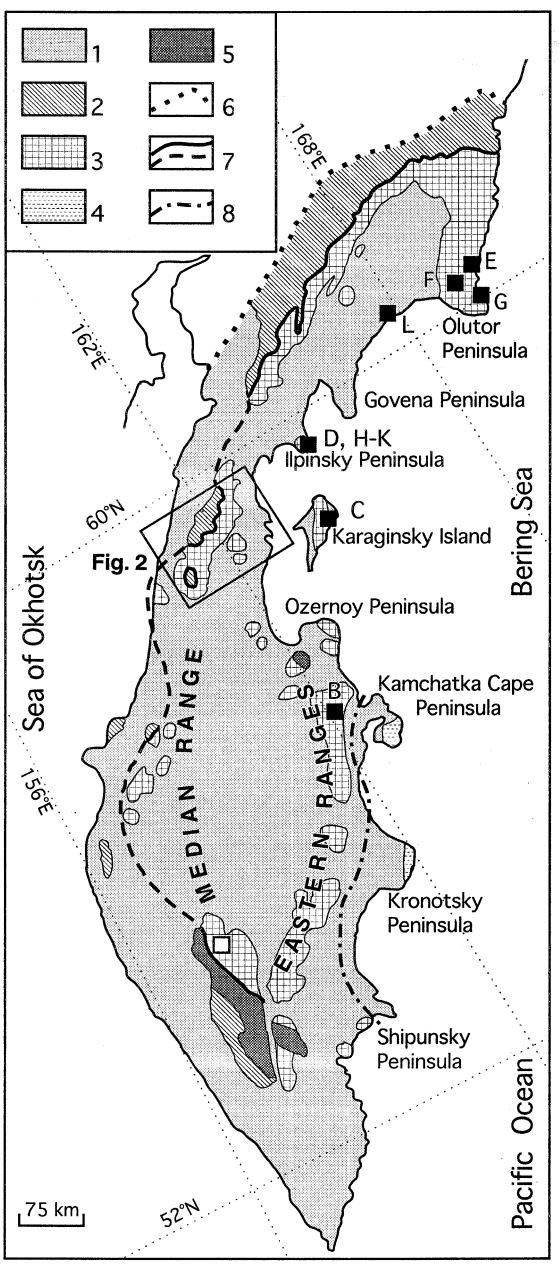


Fig. 1. Distribution of pre-Cenozoic complexes of Kamchatka and the southern part of the Koryak Highlands: *1* = Cenozoic cover; *2–4* = Late Cretaceous complexes: *2* = terrigeneous formations of the West zone; *3* = volcanoclastic formations of the Median Range and East Ranges zones and the Olutor zone of the Koryak Highlands; *4* = East Peninsulas zone; *5* = pre-Cretaceous metamorphic complexes; *6* = the northern boundary of the Olutor–Kamchatka realm; *7* = the Lesnov thrust shown as a thick solid (thick dashed) line where it is mapped (inferred); *8* = eastern boundary of the East Ranges unit. Sampling locality in the southern part of the Median Range is shown as an open square. The areas where paleomagnetic data are available (solid squares) are labeled as in Table 2.

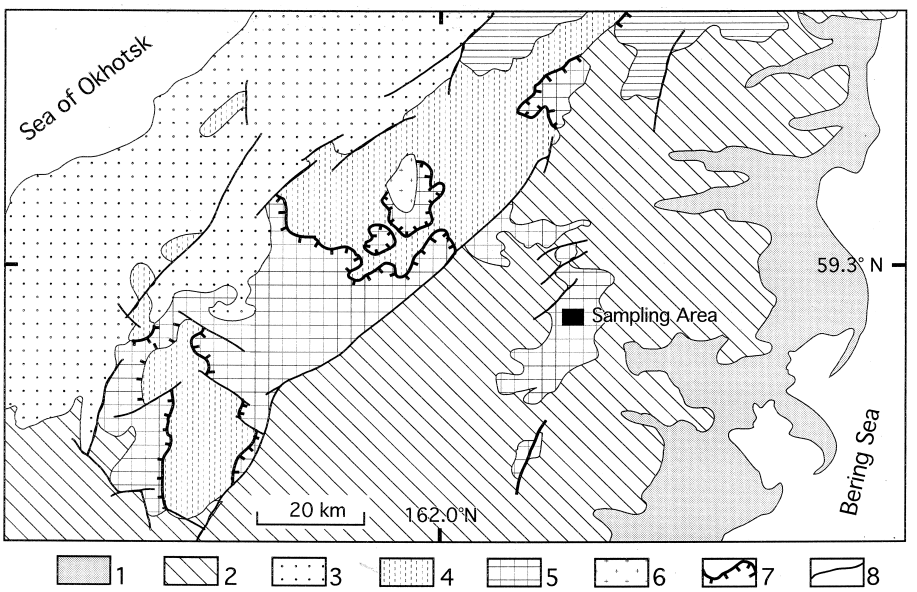


Fig. 2. Schematic geological map of the Lesnov Uplift in the northern part of Kamchatka: *1* = Quaternary rocks; *2* = Upper Eocene, Oligocene and Neogene sedimentary and volcanic rocks; *3* = Paleocene–middle Eocene sedimentary and volcanic rocks; *4–5* = Upper Cretaceous: *4* = Lesnov Series; *5* = Iruney Formation; *6* = Shamankin granite massive; *7* = Lesnov thrust; *8* = other faults. Sampling locality shown as a solid square.

this model, a Santonian–Campanian island-arc complex (Iruney Formation) in the northern part of the Median Range (Fig. 2) and the Upper Cretaceous–lower Paleocene island-arc related Iruney and Kirganik Formations in the southern part of the Median Range (Fig. 1, separate map is not given) were sampled, and the results obtained on these rocks are presented in this paper.

2. Geological setting and sampling

Late Cretaceous–early Paleocene arc-related complexes are exposed mainly in the NNE–SSW trending Median Range and the East Ranges tectonic zones of Kamchatka. The Median Range complex is thrust over the West zone of Kamchatka along the large gently dipping Lesnov thrust of north–west vergence. In the northern part of the Median Range (Fig. 2), the autochthon of the Lesnov thrust comprises turbidites and contourites of the Lesnov Series which have been interpreted to represent passive margin continental slope facies [10]. This series is strongly deformed, and a tectonic melange is present along the thrust front. A late Cretaceous age of the Lesnov Series is deduced from rare inoceram fragments [11]. The composition and structural position of this series are analogous to those of terrigenous flysch of the Ukelayat zone of the Koryak Highlands; this flysch contains Campanian–Maastrichtian radiolaria and Paleocene–early Eocene foraminifera [4,12]. Furthermore, in the West zone of Kamchatka, lithological counterparts of the Lesnov Series greatly differ from overlying late Paleocene coal-bearing sediments of continental provenance [13]. Thus, the accumulation of the Lesnov Series encompasses the Campanian, Maastrichtian, and, probably, the very beginning of the Paleocene.

The allochthon of the Lesnov thrust consists of several nappes. All thrust sheets comprise mainly siliceous and tuffaceous rocks and basalts of the Iruney Formation, but the section composition varies considerably from one sheet to another. On the western slopes of the Median Range, the structurally lowest thrust sheets are exposed. Here the Iruney Formation includes layers of red jasper and oceanic pillow-lava with greenish-gray siliceous tuffaceous rocks at the section's top. Santonian–Campanian

inocerams were found here [11,14]. On the eastern slopes of the Median Range, structurally higher thrust sheets consist mainly of gray siliceous and tuffaceous rocks with inocerams and their debris at the section base, whereas basalts and tuffs prevail at the section's top. The allochthonous complex is overlain with angular and erosional unconformities by Cenozoic sediments and volcanics, the volcanics of the Kinkil Formation being the oldest. Their age is poorly defined, but these volcanics are overlain by the Middle–Upper Eocene marine sediments [13] and thus certainly accumulated before the late Eocene.

The lower age limit of the Lesnov thrust is constrained by the age of the youngest rocks of its autochthon which is close to the end of the Cretaceous, ca. 65 Ma. Its upper age limit is constrained by the oldest overlying rocks which are middle Eocene or older. Besides, the thrust is intruded by the granites of 45 Ma age [15]. Therefore, the activity of the Lesnov thrust is limited to the 65–45 Ma interval. Other events, however, took place after major thrusting and folding. There was deep erosion after thrusting, then the volcanics of the Kinkil Formation accumulated, and, finally, deep erosion occurred again before the beginning of the late Eocene. These processes must have take some time. It seems to be logical, therefore, to assume that the main thrusting along the Lesnov thrust occurred no later than the beginning of Middle Eocene time or even earlier, perhaps the beginning of Eocene. Thus we can suppose that the upper limit on the Lesnov thrust activity is probably about 55–50 Ma.

The Late Cretaceous island-arc complexes were sampled in the northern and southern parts of the Median Range (Fig. 1). The northern locality is located east of the Lesnov Uplift (Fig. 2). On the eastern slopes of the Median Range, the Late Cretaceous island-arc complex is covered by Miocene–Pliocene volcanics, and Cretaceous rocks outcrop only where the young volcanics are deeply eroded. Here the Iruney Formation is deformed into numerous NNW-trending gentle folds. At the sampling locality, the section base is a 120-m-thick member of siliceous–tuffaceous rocks with subordinate acid tuffs and numerous lenses filled with inoceram fragments. This member is conformably overlain with gradual transition by more than 100 m thick black tuffaceous siltstones with interbeds of psammitic and gravel tuffs. Lenses of siliceous and tuffaceous rocks with abundant fragments of inoceram shells are typical for the Iruney Formation in the northern Kamchatka where these rocks are considered to be Santonian–Campanian in age. There are no direct data on the age of the sampled section, and we have to assume that it is of the same age as the Iruney Formation in the other parts of Kamchatka. Seventy four oriented samples were taken at six sites from beds with various attitudes.

The Late Cretaceous–early Paleocene island-arc complex is divided into the Santonian–Campanian Iruney and Maastrichtian–Danian Kirganic Formations in the south of the Median Range (Fig. 1, separate map is not given). Sixty samples from both tuffs, siliceous and siliceous–tuffaceous rocks of the Iruney Formation and tuffs, tuffaceous sandstones and siltstones of the Kirganik Formation were sampled at five sites.

The true thickness of the studied outcrops, which are treated as sites here, varies from several tens of meters to more than 100 meters (Table 1). One hand-sample oriented with a magnetic compass was taken from each stratigraphic level, samples being spaced more or less uniformly across the sampled outcrops.

3. Methods

One to three, usually two, cubic specimens from each hand-sample were subjected to thermal demagnetization in 12–18 steps up to 580ºC. One specimen from each sample was thermally demagnetized in a home-made oven with internal residual fields of about 10 nT and measured on a JR-4 spinner magnetometer with a noise level of 0.05 mA/m in the Paleomagnetic Laboratory of Geological Institute in Moscow. Another sister specimen was thermally demagnetized in a custom-made furnace and measured on a cryogenic magnetometer in the magnetically shielded room in the Paleomagnetic Laboratory of the Institute of Physics of the Earth in Paris. Demagnetization results were plotted on orthogonal vector diagrams [16], and linear trajectories were used to determine directions of magnetic components by a least square fit comprising three mea-

Table 1
Paleomagnetic data from the Upper Cretaceous Iruney Formation

Site	N_0/CD/RC	Th (m)	A/d	In situ				Tilt corrected			
				D (°)	I (°)	K	α_{95} (°)	D (°)	I (°)	K	α_{95} (°)
1	8/4/3	30	44/31	72.8	−35.8	12.6	17.9	98.0	−61.2	27.3	11.9
2	9/3/3	40	27/31	71.5	−50.4	70.3	8.0	118.1	−64.0	41.5	10.5
3	14/4/6	85	35/27	60.9	−54.7	17.6	12.8	104.8	−74.5	18.2	11.8
4	10/3/4	50	318/38	33.7	−69.5	47.6	8.9	100.8	−54.0	38.9	9.9
5	10/5/2	70	326/27	21.2	−68.5	40.3	9.7	94.1	−66.7	30.7	11.2
6	23/4/8	110	323/38	336.6	−59.7	26.9	8.7	115.2	−74.8	41.6	6.9
Mean (samples)	74/23/26			44.4	−65.0	10.0	6.8	105.5	−67.0	25.2	4.1
Mean (sites)	6/6			48.7	−61.3	14.3	15.2	104.2	−66.1	83.3	6.3
$F_{(2,8)} = 4.46$				$f = 16.86$				$f = 0.10$			

Comment. Site, site number; N/CD/RC, number of samples studied/component directions/remagnetization circles; Th, stratigraphic thickness studied; A/d, azimuth of dip/dip angle of bedding; D, declination; I, inclination; K, Fisher concentration parameter; α_{95}, radii of 95% confidence circle; F = 95% critical value of F statistic with the numbers in parentheses denoting degrees of freedom; f = calculated values of F statistic. The underlined result was used for interpretation.

surements or more [17]. Paleomagnetic components isolated from sister specimens were used to calculate sample-means which were in turn used to calculate site-means. Paleomagnetic software written by Randy Enkin and Stanislav V. Shipunov for the IBM PC and by Jean-Pascal Cogné for the Macintosh was used in the analysis.

4. Paleomagnetic results

Apart from an unstable remanence removed below 300–350°C, no consistent directions of magnetization were isolated in the Upper Cretaceous rocks from the southern locality, and the entire collection was discarded. In contrast, siliceous–tuffaceous rocks from the northern locality yielded much better data. Intensities of natural remanent magnetization (NRM) of these samples range from 0.07 to 300 mA/m. A significant abatement of magnetization intensity below 200° to 350° is most likely due to removal of an overprint component of recent origin. This low temperature component is highly scattered, and its mean direction is close to that of the present-day field. A characteristic remanent magnetization (ChRM) is identified in some samples by linear decay to the origin above 200° to 350°C (Fig. 3a,b). In many samples, however, the ChRM is isolated at much higher temperatures, and either linear (Fig. 3c) or, more often, curved (Fig. 3d) segments of the trajectory can be observed between 300° and 480°C. These linear segments apparently represent an intermediate temperature component. The directions of this component, however, are very scattered both in situ and in stratigraphic coordinates and generally fall between the overprint direction and that of the ChRM. We attribute this pattern to partial overlap of unblocking temperature spectra of the recent overprint and the ChRM. In still other samples, the ChRM could not be identified (Fig. 3e), and only remagnetization circles were defined. Finally, many samples showed erratic behavior above 400–430°C, and this behavior was the main reason for data rejection.

All ChRM directions are of reversed polarity and are well grouped at each site. Site-mean directions were defined either using only the ChRM directions, or combining the ChRM directions with remagnetization circles [18]. Sections of fine-grained tuffaceous rocks, treated as sites here, are 30 to 110 m thick (Table 1), and we assume that the accumulation of each section had taken a rather long time interval. The calculated site-means are well defined (Table 1). The fold test [19] is positive at the 95% confidence level (Table 1), indicating prefolding age for the ChRM. The best data grouping is reached at 90% unfolding, the maximum differing by just two percent from the tilt-corrected value of concentration

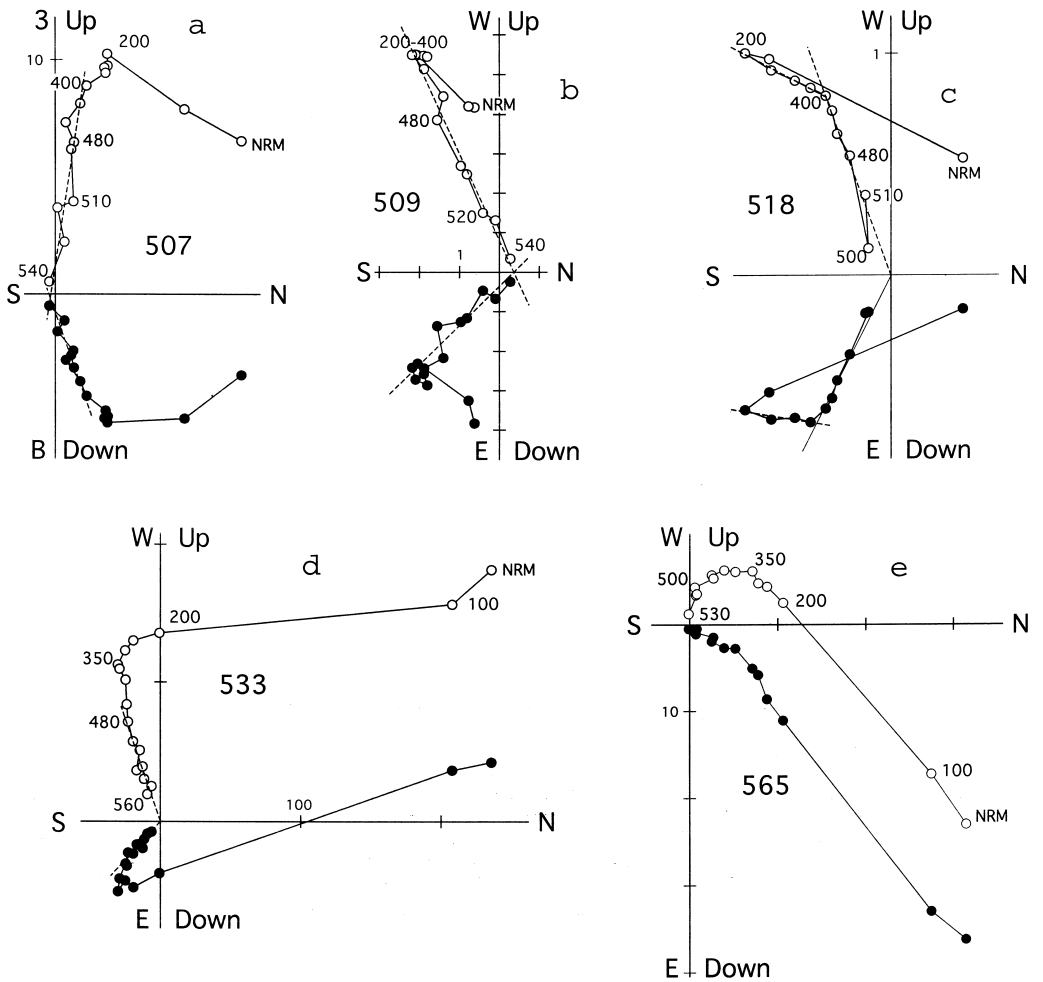


Fig. 3. Representative vector component plots for Campanian tuffaceous rocks of the Iruney Formation. Solid (open) symbols are projected onto the horizontal (vertical) plane. The plots are in stratigraphic coordinates. Isolated components are shown as dashed lines. Temperatures are in degrees Celsius. Magnetizations are in mA/m.

parameter. Relatively thick stratigraphic intervals studied at each site and tight grouping of tilt-corrected site-means imply that paleosecular variation is adequately averaged (Fig. 4, Table 1). Folding in the study area took place in the Paleocene–Eocene, and hence the prefolding ChRM in these upper Cretaceous rocks is likely to be primary. Maximum unblocking temperatures for all samples are between 530° and 570°C (Fig. 3) indicating that magnetite is the main remanence carrier.

The tilt corrected overall mean direction of dec. $= 105.5°$, inc. $= -67.0°$, $\alpha_{95} = 4.1°$ corresponds to a paleolatitude of $49.7° \pm 5.6°$ N.

5. Age of the studied rocks

Judging by inocerams, the studied rocks are of Santonian–Campanian age [11,14]. Paleomagnetic study shows that all ChRM directions are of reversed polarity. The Cretaceous superchron of normal polarity ended by the beginning of the Campanian, ca. 83 Ma [20]. Therefore, the studied rocks are not older than Campanian. According to the same paper, there were only one chron of reversed (83–79 Ma) and one of normal polarity (79–74 Ma) in the Campanian. Thus the accumulation of the studied rocks is constrained to the 83–79 Ma interval.

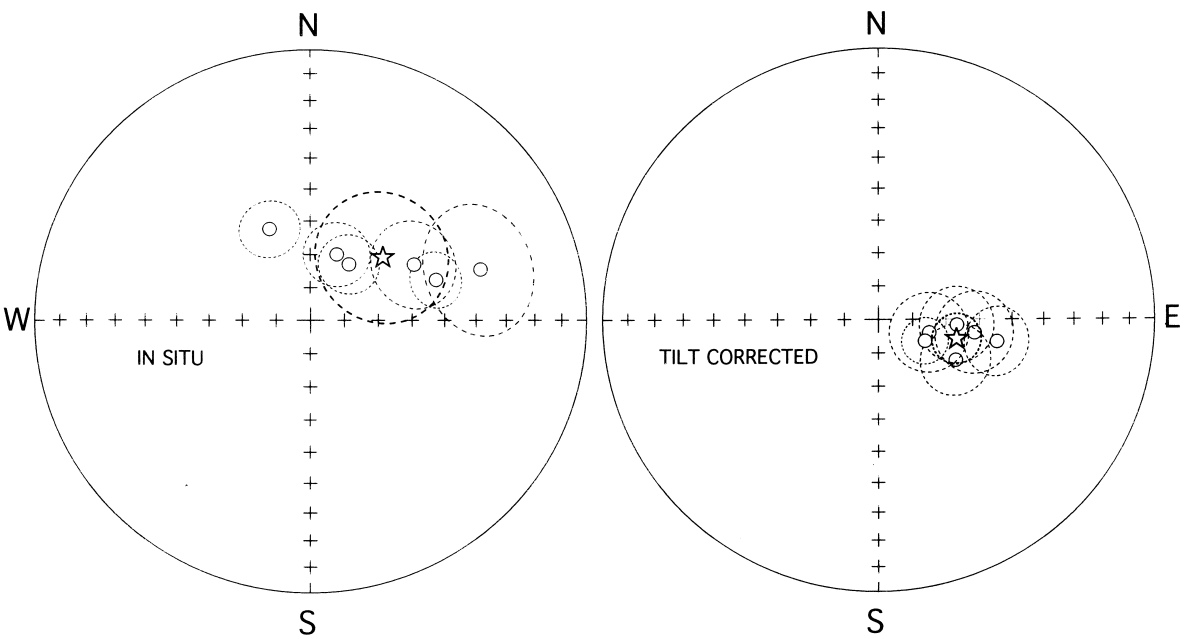


Fig. 4. Stereoplots of ChRM site-mean directions (dots) with associated confidence circles (thin dashed lines) and overall mean directions (star) with its confidence circles (thick dashed line) in situ (left) and after tilt-correction (right). All symbols are projected onto the upper hemisphere.

6. Paleomagnetic data on the Late Cretaceous island-arc complexes of Kamchatka and the Koryak Highlands

In addition to our result from the northern part of the Median Range (paleolatitude 49.7° ± 5.6°N), paleomagnetic data are available from other fragments of the Campanian–Maastrichtian island arc/arcs (Fig. 1). The Late Cretaceous island-arc complex with an age of at 83–79 Ma from the East Ranges zone (Kumroch Range) accumulated at a latitude of 48.7° ± 5.0°N (Table 2, B) [6]. This result is confirmed by the fold and conglomerate tests. The Campanian–Danian island-arc complex of Karaginsky Island at the northern continuation of the East Ranges zone accumulated at 44.5° ± 5.6°N (Table 2,C) [7]. From the southern part of the Koryak Highlands, paleomagnetic data are available on Campanian–Maastrichtian island-arc complexes of the Ilpinsky Peninsula (42.1° ± 7.1°N, (Table 2,D)) [8] and Olutor zone (49.0° ± 10.9°N; 47.0° ± 6.5°N; 51.1° ± 7.0°N (Table 2,E–G)) [9,21]. All these results are based on fully demagnetized data and confirmed by the fold test. Besides, the results from the objects D, E, F, H and I are confirmed by the reversal test. The paleolatitudes derived from the studied Late Cretaceous island-arc complexes coincide within the errors of the method (Table 2).

7. Interpretation and discussion

We calculated the expected latitudes of the study area assuming that it belonged to either the Eurasian or North American plate. All calculations were made in the hot-spot reference frame, using kinematic parameters from Engebretson et al. [22]. The expected Campanian latitudes of about 69–71°N and 70–72°N for the Eurasian and North American plates, respectively, are very similar. Northeast Asia is more likely to have been a part of the North American plate [23–25], and the following discussion will be limited to this case.

All observed Late Cretaceous paleolatitudes agree within error limits, and we assume that most arc-related complexes of the Koryak–Kamchatka region had originated within the same paleo-island arc which is named the Achayvayam–Valagina is-

Table 2
Paleomagnetic data on the island-arc complexes of the Koryak–Kamchatka region

N	RA	Localities coordinates		Observed directions				Lat_{exp} (°)	Lat_{obs} (°)	ΔLat (°)	Ref.
		Lat (°N)	Lon (°E)	*D* (°)	*I* (°)	*K*	α_{95} (°)				
Upper Cretaceous rocks											
A	K_2^{cp}	59.3	162.1	105	−67	25	4.1	70–72	49.7 ± 5.6	20–22	1
B	K_2^{cp-ma}	56.5	162.0	160	−68	16	3.7	68–70	48.7 ± 5.0	19–21	2
C	K_2^{cp-ma}	59.0	164.4	332	63	20	4.5	69–73	44.5 ± 5.6	25–29	6
D	K_2^{cp-ma}	59.8	164.9	299	61	18	6	70–74	42.1 ± 7.1	29–32	5
E	K_2^{cp-ma}	61.4	171.5	80	66	15	8	72–75	49.0 ± 10.9	23–26	4
F	K_2^{cp-ma}	60.9	171.5	88	65	14	5	72–75	47.0 ± 6.5	25–28	4
G	K_2^{cp-ma}	60.9	171.7	97	68	12	5	72–75	51.1 ± 7.0	21–24	4
Paleogene rocks											
H	Tp^{Th}	59.8	164.9	285	75	12	8	67–68	61.8 ± 13.2	5–6	5
I	Te^{Ypr}	59.8	164.9	321	76	14	6	65–66	63.5 ± 10.2	2–3	5
J	$Te^{Lut–Brt}$	59.8	164.9	299	73	10	10	63–65	58.6 ± 15.8	4–6	5
K	Te^{Prb}–To^{Rup}	59.8	164.9	338	80	10	9	62–63	70.6 ± 16.3	−8–9	5
L	Te	60.6	168.0	348	74	17	4.2	64–67	60.2 ± 6.8	4–7	3

Comment. *N*, sampling locality labeled as at Fig. 1. RA, rock age: K_2^{cp-ma}, Upper Cretaceous, Campanian to Maastrichtian; Tp^{Th}, Paleocene, Thanetian; Te^{Ypr}, Eocene, Ypresian; $Te^{Lut–Brt}$, Lutetian to Bartonian; Te^{Prb}–To^{Rup}, Priabonian to Oligocene, Rupetian. Localities coordinates are from original papers. Observed data are in stratigraphyc coordinate system: *D*, declination; *I*, inclination; *K*, Fisher concentration parameter; α_{95}, radii of 95% confidence circle. Lat_{exp}, Lat_{obs}, the expected and observed paleolatitudes for the studied localities; ΔLat, the difference between the expected and observed paleolatitudes. Ref, references: 1 = this study; 2 = Kumroch Range [6]; 3 = Malinovsky Range [21]; 4 = Olutor zone [9]; 5 = Ilpinsky peninsula [8]; 6 = Karaginsky island [7].

land arc, AVIA, after Zinkevich and Tzukanov [2]. The observed paleolatitudes are 20–25° lower than expected (Table 2, Fig. 6), and northward transportation of the AVIA is strongly indicated. Volcanic activity originated at the beginning of the Campanian and lasted until the beginning of the Paleocene during which this island arc marked a convergent boundary between the North American plate and an oceanic plate. The AVIA could have been either separated from the continental margin by oceanic crust or located near to the continental margin. Thus the northward drift of the island arc could have resulted from either displacement with an oceanic plate (or plates) before collision with the continental margin, and/or movement along strike-slip faults, as it well established, for instance, for the western margin of North America (e.g., [26,27]).

In making a correct kinematic model, one should know the paleolatitudes and period of volcanic activity for an island arc and the time of its arrival at the continental margin. The latter can be constrained by the age of collision-related folding and thrusting; in our case, it is the age of thrusting along the Lesnov thrust. The activity of this thrust is broadly confined to the 65–45 Ma interval; it has already been shown that the major activity along it had stopped somewhat earlier, most likely before 55–50 Ma. Thus, the docking time for the AVIA is limited to the interval between 65 and 55, less likely as late as 45 Ma. Therefore, northward drift of the island arc could have lasted at most from the beginning of Campanian until 45 Ma, which is the latest possible docking time, but probably stopped somewhat earlier. It is still unclear, however, where the Kula–Pacific transform boundary was located in the Late Cretaceous–early Paleocene, and scenarios of coastwise translation and intraoceanic transport are analyzed for both oceanic plates.

Geochemically, the volcanoclastic complexes of the Achayvayam–Valagina island arc are similar to volcanics of intraoceanic island arcs [4,5], and it could hardly have been coast-parallel transported during the time of volcanic activity in the Late Cretaceous–early Paleocene. Nevertheless, the maximal coastal translation of the studied island arc for the 83–45 Ma interval was calculated for both the

Kula and the Pacific plates. The subducting and overriding plates interact in a complex manner and the true coast-parallel velocity should be less than the tangential component of the total velocity [28], most probably not exceeding 60% of it [29]. The calculation showed that the maximal coastal translation is about 5–6º and 6–7º of latitude for the Kula and Pacific plates, respectively. Thus, even if the coastwise translation had taken place, it can account for only a minor part of the northward displacement of the Achayvayam–Valagina island arc.

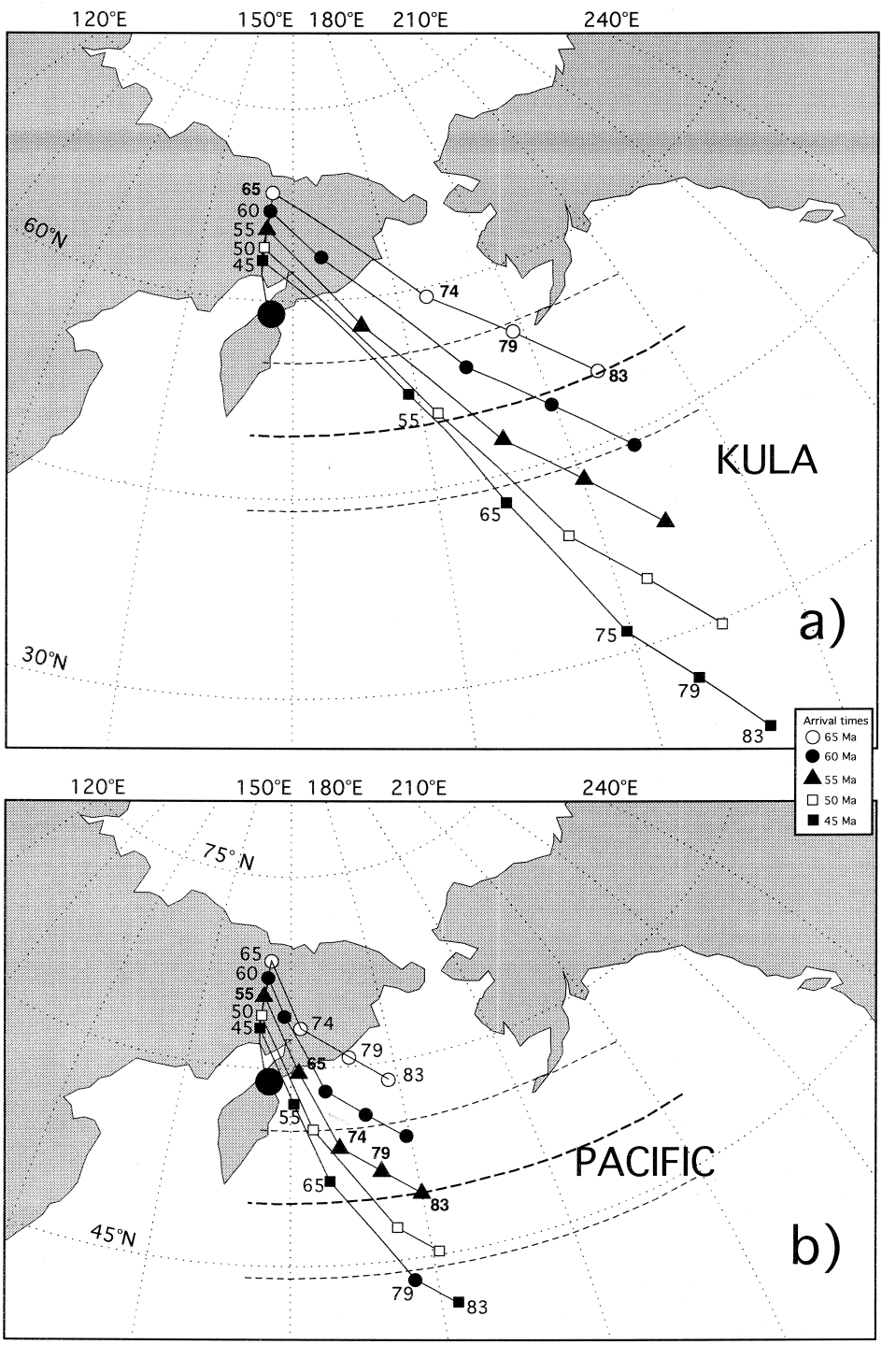
As considerable coastwise transport has been ruled out, we next assume that the AVIA was drifting northwestward with one of the oceanic plates. Following the method of Debiche et al. [30], the trajectories for the Kula and Pacific plates as the carrier plate were backward modeled as illustrated for the studied part of the Median Range (Fig. 5). Calculations were made for the age of the studied complex of 83–79 Ma and docking times which vary between 65 and 45 Ma. Motion with the Kula plate predicts an arrival time of about 65–60 Ma (Fig. 5a), whereas motion with the Pacific plate predicts an arrival time of about 55–50 Ma (Fig. 5b).

The docking time for the AVIA is not well enough known to discriminate between these two scenarios. However, we have some information which can additionally constrain the docking time of this arc. In the East Ranges, a gradual transition from Upper Cretaceous island arc volcanoclastics to the overlying lower Paleocene continent-derived turbidites is observed. This flysch, though not a proximal deposit sensu stricto, accumulated close to the continental slope [10], and the island arc was unlikely to have been more than a few hundred kilometers away from the provenance area at about 60–65 Ma [6]. This means that the arc had already moved most of its way north by the beginning of Paleocene (65–60 Ma) even if the collision with the continent occurred later. Also, paleomagnetic data on Cenozoic rocks from the Olutor zone of the Koryak Highland [8,21] show that the Paleocene–Oligocene complexes were formed at paleolatitudes similar to expected ones for the North American plate (Table 2, H–L). Therefore, it seems reasonable to assume that the studied island arc moved with the Kula plate and arrived at the continental margin in the early Paleocene (Fig. 5a).

Northward drift of the AVIA coincides with its activity. This implies that the related subduction zone was dipping under the Kula plate and had to move in the absolute reference frame with a rapidly moving oceanic plate. Most likely this subduction zone was consuming the oceanic periphery of the North American plate. The cessation of volcanic activity coincided with the end of northward drift of the paleoarc in the beginning of the Paleocene. We assume that the cessation of volcanic activity of the Achayvayam–Valagina island arc and slowing down of its drift in the early Paleocene resulted from its collision with the continental margin. On the other hand, the formation of the suture zone between the arc and margin and deformation in this zone are likely to have started later. On Kamchatka, deformation took place in the early Eocene, while it is of Miocene age in the Olutor zone of Koryakia. This diachronous deformation may be accounted for by oblique arc–continent collision that had started in southwest Kamchatka. As a result, motion of the Kula plate was possibly slowed down and its boundaries were re-arranged. Simultaneously, the new Aleutian and Kamchatka subduction zones are likely to have been formed.

So far, our interpretation has been confined to the observed latitudes. The AVIA motion on the Kula plate resulted in less than 10º declination anomaly, whereas most paleomagnetic declinations, except for the data from the Olutor peninsula (localities E–G, Fig. 1, Table 2), are much more deflected westward with respect to North America (Table 2) indicating counterclockwise rotations of different parts of the AVIA. This pattern is broadly compatible with the assumption that the arc–continent collision started in the south, and that the entire island arc was later rotated and deformed. On the other hand, this pattern may also arise from post-collisional sinistral strike-slip motion of the AVIA as the result of minor coast-parallel transport. At this state of art, both scenarios appear plausible.

In conclusion, we stress that the paleomagnetic data, the ages of the rocks, and kinematic parameters of the oceanic plates used in our model are not accurate enough to ensure that the proposed scenario is entirely correct. However, we can confidently state that the Achayvayam–Valagina island arc was formed on the oceanic crust 20–25º south of the continental margin in the Campanian (Fig. 6). It


120°E
150°E
180°E
210°E
240°E
60°N
30°N
65
60
55
50
45
74
79
83
55
65
75
79
83
KULA
a)
Arrival times
65 Ma
60 Ma
55 Ma
50 Ma
45 Ma
75°N
45°N
PACIFIC
b)

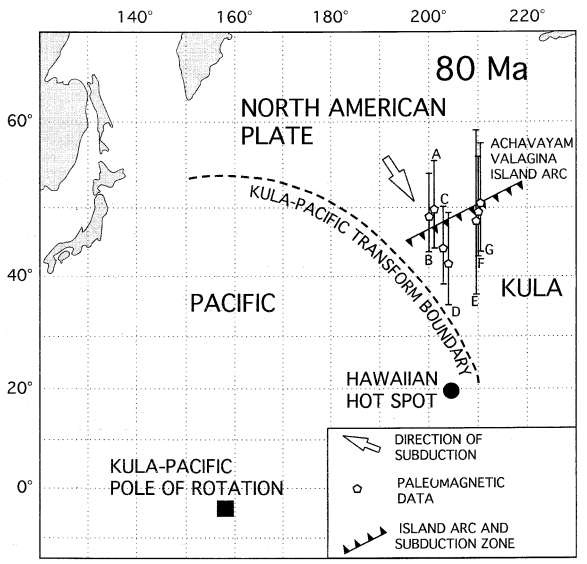


Fig. 6. Reconstruction of the northwest Pacific and its margins at 80 Ma in the hot-spot reference frame (simplified after Engebretson et al. [22]). The Achayvayam–Valagina island arc and the related subduction zone (toothed line) are moving with the Kula plate. Oceanic periphery of the North American plate is subducting (open arrow) under the Kula plate. Paleolatitudes derived from different parts of the Achayvayam–Valagina island arc are shown as open pentagons and labeled as in Fig. 1 and Table 2; error bars are shown as vertical lines.

was transported northward by an oceanic plate, most likely the Kula plate, until arrival at the continental margin in the early Tertiary. More reliable paleomagnetic data on well-dated rocks are needed to further refine the tectonic evolution of Kamchatka.

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Fig. 5. Trajectories of the studied fragment of the Achayvayam–Valagina island arc with (a) the Kula plate and (b) the Pacific plate for arrival times from 65 to 45 Ma. The observed paleolatitude and its confidence limits are shown as thick and thin dashed lines, respectively. The trajectories (solid lines) are backward modeled [30] in the hot-spot reference frame using the kinematic parameters of Engebretson et al. [22]. Kamchatka is assumed to belong to the North American plate. Eurasia and North America are shown in their present-day positions. Large solid dot is the study area.

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